# Nemanja Mikić

**Senior Software Engineer · Backend, Platform & Cloud · Java, Python, Go**

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## Professional Summary

Software engineer with 8+ years of experience building backend services, developer frameworks and cloud integrations in Java, Python and Go. At Oracle since 2022 (IC4) with 400+ merged pull requests across the Micronaut ecosystem: OpenTelemetry distributed tracing, vector search in Micronaut Data, Oracle Cloud (OCI) and AWS integrations, Kubernetes workload identity and GraalPy Python/Java interoperability. Author of official Micronaut Guides on Kubernetes and observability; credited in Apple's web server security acknowledgements (November 2025). MSc in Software Engineering, University of Belgrade.

## Core Skills

**Languages:** Java, Python, Go, SQL

**Backend and frameworks:** Micronaut, Netty, GraalVM native image, GraalPy, REST and gRPC APIs, microservices, JDBC, R2DBC, Hibernate, JDBI, Jersey, Django, Tornado, SQLAlchemy

**Cloud and infrastructure:** Oracle Cloud Infrastructure (OCI), AWS, Google Cloud (GCP), Kubernetes (OKE, EKS, GKE), Docker, Terraform, infrastructure as code, Oracle Functions

**Delivery and DevOps:** CI/CD, GitHub Actions, GitLab CI, Gradle, Maven, release automation, dependency management, smoke and integration testing, Git

**Observability:** OpenTelemetry, Micrometer, Prometheus, Amazon CloudWatch, distributed tracing, metrics, logging

**Data:** PostgreSQL, MySQL, Oracle Database, MongoDB, vector and similarity search (pgvector, MySQL HeatWave)

**Security:** Application security, penetration testing, OAuth2, OpenID Connect (OIDC), SSO, MFA, TOTP, HashiCorp Vault, TLS and certificate management

## Experience

### Oracle — Software Engineer (IC4)

Jan 2022 – Present

IC4 since June 2024; IC3 from January 2022 to May 2024 · Micronaut framework and GraalVM ecosystem · Belgrade, Serbia

- 400+ merged pull requests across the Micronaut ecosystem since 2022, spanning framework features, cloud integrations, observability and developer tooling.

**Observability and framework integrations**

- Implemented Micronaut OpenTelemetry support across HTTP and gRPC clients and servers: span lifecycle handling, tracing annotations, SDK configuration, tests and documentation.
- Added JDBC and R2DBC tracing, Micrometer Observation and Prometheus PushGateway integrations, with configuration support, automated tests and documentation.

**Cloud SDKs and service reliability**

- Built AWS CloudWatch logging and OCI integrations for cloud logging, certificate management and reloading, OKE workload identity and configurable signing-proxy requests.
- Fixed synchronous OCI retry back-off that could run on Netty event loops and stall server I/O; added regression tests for responsive health checks, retry behaviour and exception preservation.

**Micronaut Data vector search**

- Implemented dense and sparse vector types, derived similarity queries, scoring and schema generation, with Oracle and PostgreSQL vector indexes (PR #3637, documentation).
- Added JDBC integrations for Oracle Database, PostgreSQL (pgvector) and MySQL HeatWave, plus Oracle and PostgreSQL R2DBC, with Java, Kotlin and Groovy tests and examples.

**Python interoperability and native developer tooling**

- Improved GraalPy-based Python/Java interoperability in Micronaut: coroutine/Publisher adaptation and automatic dataclass introspection with typed validation annotations.
- Extended Micronaut Python project generation with service discovery, and Pyronaut tooling with async pytest, PyPI releases, multi-architecture containers and Windows PGO.

**Published guides and sample applications**

- Authored Micronaut Guides on Kubernetes (Oracle OKE, Amazon EKS, Google GKE) and metrics (AWS CloudWatch, Oracle Cloud); co-authored OCI OpenTelemetry tracing.
- Added tested Python OAuth2/OIDC samples for Auth0, Keycloak and Amazon Cognito, plus CORS, token propagation, async HTTP clients and Oracle Database integrations.

**Delivery automation and integration testing**

- Developed scheduled dependency-update and branch-synchronisation tooling that opens review pull requests across Micronaut service and reference repositories, with OCI resource-principal authentication for repository access.
- Implemented release pipelines with ticket checks, canary and template validation, tag promotion and rollback; added Maven/Gradle smoke tests and offline dependency-mirror validation with checksum and compliance reports.

**Multi-cloud application tooling**

- Extended Graal Development Kit application and Kubernetes generation across AWS, OCI and GCP with cloud configuration, metrics/tracing and generated tests; contributed to an internal Micronaut service framework with metrics, request-ID propagation and Terraform references.
- Implemented Oracle Functions generation and GitLab CI stages for JDK 17/21, GraalVM native builds, guide testing, dependency scanning and artifact publishing; contributed OCI DevOps project generation and recurring versioned releases.

**Seven Bridges Genomics — Penetration Tester** May 2021 – Dec 2021

- Moved to the security team to test the platform I had helped build: web and API penetration testing, authentication and authorization reviews, and remediation guidance for the Core Team.

**Seven Bridges Genomics — Software Engineer, Core Team** Dec 2017 – May 2021

Platform infrastructure and backend services in Java, Python and Go

- Built a secrets management service in Go with HashiCorp Vault, allowing users to store sensitive credentials and organisation administrators to share secrets with authorised members.
- Implemented MFA recovery codes, email verification and time-based one-time passwords in a Python service; improved OAuth2 single sign-on flows for custom applications in Java and Python.
- Split a large billing service into smaller Java services for customer charges and platform resource pricing; developed the administrative billing API gateway and HTTP clients in Go.
- Implemented a Go service that monitors AWS EC2 test environments and shuts them down after 90 minutes without user activity.
- Developed a Go and MongoDB messaging service for administrative announcements to anonymous users, registered users or individual recipients.
- Stack: Django, Jersey, Jackson, Guice, JDBI, Hibernate and go-chi across backend services and platform integrations.

**Seven Bridges Genomics — Python Software Engineer Intern** Mar 2017 – Jun 2017

- Built a Python user management service for registration and authentication, and an API gateway for frontend clients using Tornado, SQLAlchemy and PostgreSQL; worked with REST APIs, Git and a microservices architecture.

## Education

**MSc in Software Engineering** — University of Belgrade, School of Electrical Engineering Oct 2017 – Sep 2019

**BSc in Software Engineering** — University of Belgrade, School of Electrical Engineering Oct 2013 – Sep 2017

**Languages:** Serbian (native), English (professional working proficiency)

## Recognition and Publications

- Credited in Apple's web server security acknowledgements, November 2025.
- Micronaut Guides: Service discovery and distributed configuration with OCI OKE; with Amazon EKS; with Google GKE; Collect Micronaut metrics with Amazon CloudWatch; Collect Micronaut metrics on Oracle Cloud; OpenTelemetry tracing with Oracle Cloud (co-author).